

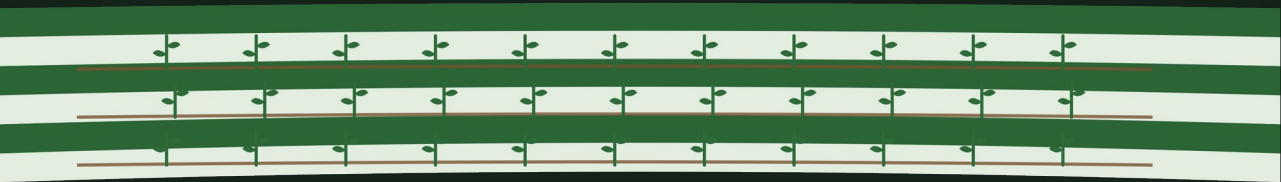
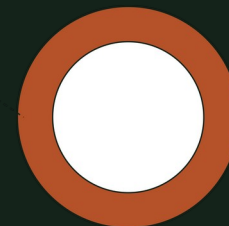
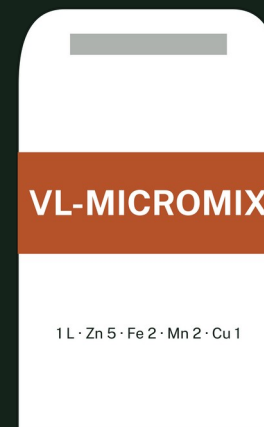


DEALER TRAINING · SEPTEMBER 2026

VL-Micromix

4 micronutrients in 1 litre, sprayed on the leaf at 1 litre an acre, at the stage the programme names.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

VL-Micromix is 4 micronutrients in 1 litre: zinc 5%, iron 2%, manganese 2% and copper 1%, weight in volume, so 50 g of zinc, 20 g of iron, 20 g of manganese and 10 g of copper in the bottle, fully soluble. A crop eats these in grams an acre and cannot finish without them: zinc makes the growth hormone, iron and manganese run the leaf, copper builds the stem and the pollen. On calcareous ground above pH 8 the soil locks all 4, and 59.0% of Punjab samples already read low in zinc. So this bottle goes on the leaf, where the soil never gets a vote, 1 litre an acre in 100 to 200 litres of water, at early growth on wheat and rice, and at the rows the plan prints for potato, chili, pulses and guava. It corrects the small nutrients. The big 3 are the bags.

The 4 things it must contain

1

The bottle: 1 L, Zn 5 · Fe 2 · Mn 2 · Cu 1 (% w/v), PS 5469-2020. 50 g of zinc in every litre. A 20 L pack too.

2

The nutrients: 4 metals the crop eats in grams. None builds tissue; all 4 run the machinery that does.

3

The soil: above pH 8 the soil locks them. 59.0% of Punjab samples read low in zinc. So the route is the leaf.

4

The method: 1 litre an acre, water first, on the leaf, at the row the plan prints. Never in the phosphate tank.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

THE BAG

What the bottle says, and what it is registered as.

1 L

the bottle, 1 litre an acre in every programme row. 50 g Zn, 20 g Fe, 20 g Mn, 10 g Cu in it, from the declared analysis. A 20 L pack is on the price sheet.

Zn 5%

zinc, w/v: 50 g in a litre. The metal that leads, because it is the one the Punjab survey reads low most often.

Fe 2 • Mn 2

iron and manganese, 20 g of each in a litre. The 2 metals that run the leaf, and that high pH and lime lock together.

Cu 1%

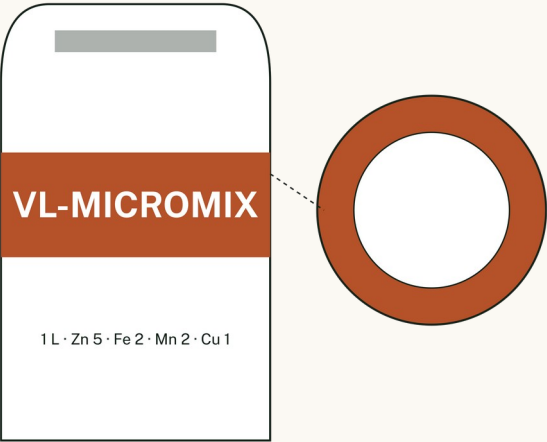
copper, 10 g in a litre. Copper reads Healthy in every Punjab district where the figure is comparable; it is in the bottle for the crop, not for a shortage VAN sells.

100%

soluble, a liquid formulation, as VAN’s product data prints it. No residue in the tank.

16

published programmes name VL-Micromix, 40 rows, 1 litre per application on every row.



Registered as	Pakistan Standard	Manufacturing licence
VL-Micromix	PS 5469-2020	CM/L-3129/2024

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

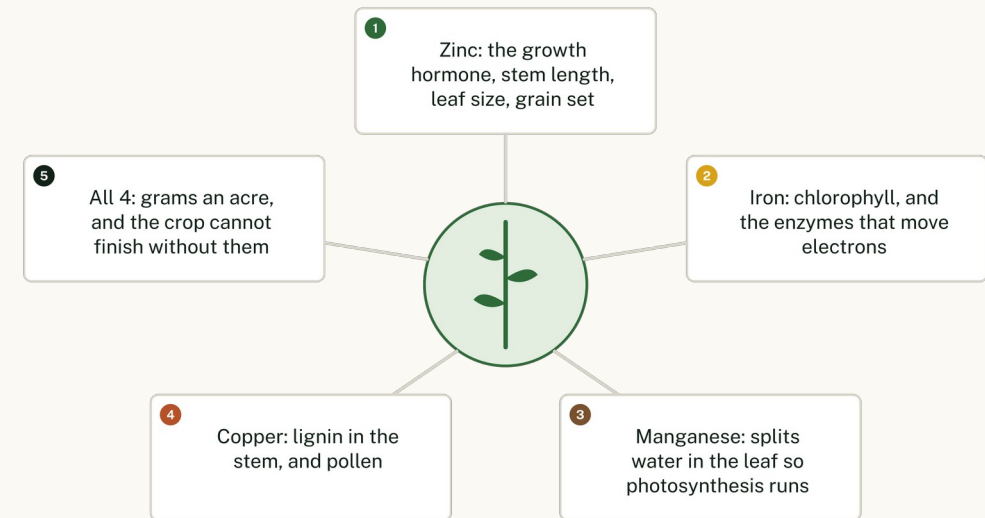
“I gave the wheat its urea and it is green, but the new leaves come out small and pale between the veins, and the plant looks bunched. What is it short of?”

The grower fed nitrogen and the crop took it. What he is describing is not a nitrogen sign: nitrogen yellows the old leaves first, from the tip back. Small new leaves, yellow between green veins, short joints: those are the small nutrients, the ones the soil locks. The answer is 3 ideas: what the 4 metals do, why calcareous soil starves the crop of them, and why the leaf is the route.

The 3 answers on the next 3 slides, 1 idea each.

4 metals, in grams, and the crop cannot finish without them.

Zinc makes the growth hormone that lengthens the stem and the enzymes that build protein; it decides plant height, leaf size and, in rice and wheat, grain set. Iron makes chlorophyll and runs the enzymes that move electrons. Manganese splits water inside the leaf so photosynthesis can run. Copper builds lignin, the woody part of the stem, and is needed for pollen to form. A field can be full of nitrogen and still stop at the zinc.

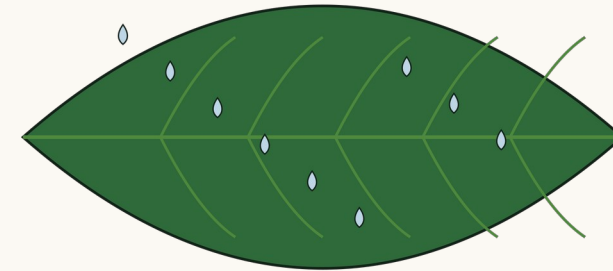


The jobs of the 4 metals, from the site's 17-nutrients page. A picture of the idea, not a measurement.

Say it like this: “These 4 do not build the plant. They run the machinery that builds it. A pinch an acre, and without it the crop stops.”

The soil locks them. The leaf does not.

On calcareous ground above pH 8, zinc is fixed as zinc phosphate or held on the lime, iron and manganese are locked by the same pH and by bicarbonate in the irrigation water, so the soil can be full of a metal the root cannot drink. 69.1% of Punjab samples read above pH 8 and 59.0% read low in zinc. A spray puts the 4 metals on the leaf, where they are taken up directly while the droplet is wet. That is the whole reason this product is a bottle and not a bag.



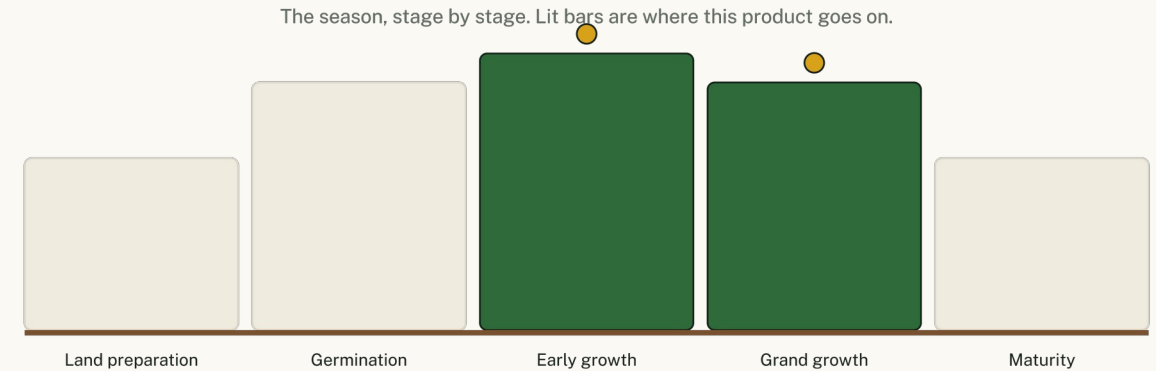
Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

Say it like this: “Your soil has the metal and locks it. The leaf takes it straight in. That is why it is a spray.”

The leaf taking up the droplet. General soil chemistry, and the Punjab survey figures; no product measurement.

Early, while the plant is still deciding its size.

The 4 metals run the enzymes of growth, so the crop asks for them while it is building leaf and setting how many tillers, joints and heads it will carry. The published programmes put VL-Micromix at early growth on wheat and rice, and through the season on potato, chili and guava, 1 litre an acre at each row. Iron does not move inside the plant once placed, so a spray at flowering is arriving for a job the plant already did without it. The sign of shortage shows in the new leaves; the loss was decided before that.



The 5 stage names are the programme's. Which rows carry VL-Micromix for your crop are on the programme slides.

Say it like this: "Spray it when the plan says, early. By the time the new leaves show it, the plant has already made itself small."

THE EVIDENCE

What is verified for VL-Micromix, and what is not.

No trial of VL-Micromix is on VAN's record. What a dealer may state as fact is the bottle, the licence, the programme rows and the Punjab soil record, each as what it is.

Item	What it says	How a dealer may use it
The bottle	VL-Micromix, Zn 5% · Fe 2% · Mn 2% · Cu 1%, w/v, liquid, 100% soluble, 1 L. A 20 L pack on the price sheet.	As fact. Grams per litre from the declared analysis.
The licence	Pakistan Standard PS 5469-2020 for liquid nutrient supplements; manufacturing licence CM/L-3129/2024, as the master record holds it.	As fact. The number, never an expiry date.
The programme rows	40 rows on 16 programmes, 1 litre per application: spray on 14 programmes; fertigation on banana (1 to 2 L) and garlic; guava prints the water, 1 L per 100 L.	As the published rate. Verbatim.
The soil record (Punjab, 770,160 samples)	59.0% of samples below 1.0 ppm zinc; mean 1.25 ppm; zinc falls as pH rises. 69.1% above pH 8. Copper reads Healthy in every comparable district (32 of 36).	As fact about the soil. Never as product performance. Copper is never sold as a shortage.
The chemistry (textbook)	Above pH 8 zinc is fixed as zinc phosphate or on lime; iron and manganese are locked by pH and bicarbonate; zinc and a soluble phosphate in 1 tank make zinc phosphate, which does not dissolve.	As general soil chemistry, not a VAN measurement.
Not on file	A VL-Micromix trial; a yield figure; the 64 / 19 / 4 / 2% national deficiency figures printed in earlier VAN material (no source named); a drone rate; a chelation claim.	Not said until named.

Sources: the bottle; VAN's published crop programmes on van.com.pk; the farmer price sheet, Aug 2026; the PSQCA master record; the Punjab soil record, 770,160 samples; VAN's 17-nutrients page; VAN's own product data.

The bottle in grams, and the 2 packs, in the grower’s own back-of-envelope.

Prices from the VAN farmer price sheet, Aug 2026, delivered farmer price. Grams from the declared analysis, w/v: 5% is 50 g in a litre.

VL-Micromix, % w/v	Zn	Fe	Mn	Cu	Rupees	Rupees per litre
1 bottle, 1 L, 1 acre	50 g	20 g	20 g	10 g	1,250	1,250
1 pack, 20 L, 20 acres	1.0 kg	400 g	400 g	200 g	17,500	875

A bottle is 100 g of the 4 metals together on 1 acre for 1,250 rupees. The 20 L pack is the same litre at 875 rupees, for a dealer, a large farm or a crop the plan sprays 5 times. The grower buys the pack his bottle count fills.

VL-Micromix 1 L PKR 1,250 and 20 L PKR 17,500: the farmer price sheet, Aug 2026. 50 g = 1,000 ml × 0.05; 20 g = 1,000 × 0.02; 10 g = 1,000 × 0.01. Rupees per litre = price ÷ litres.

What the programme puts on, per acre: the 1-spray crops.

Litres per acre from the published programmes; rupees at the VAN farmer price sheet, Aug 2026, 1 L bottles. Every row is 1 litre per application.

Crop	Stage and method, as printed	Litres	Zinc	Rupees per acre
Wheat	Early growth, spray	1	50 g	1,250
Basmati and hybrid rice	Early growth, spray	1	50 g	1,250
Chickpea, lentil, mungbean	Grand growth, spray	1	50 g	1,250
Sunflower	Grand growth, spray	1	50 g	1,250
Sesame	Early growth, spray	1	50 g	1,250
Garlic	Maturity, fertigation	1	50 g	1,250

On wheat the grower spends 1,250 rupees an acre to put 50 g of zinc and 50 g of iron, manganese and copper together on the leaf at early growth. 1 bottle, 1 pass, once. That is the counter arithmetic; what the crop does with it is not a number VAN prints for this product.

VAN's published crop programmes on van.com.pk, rows as printed. Rupees = litres × 1,250 at the 1 L price, farmer price sheet, Aug 2026.

What the programme puts on, per acre: the crops sprayed 5 times.

Litres per acre from the published programmes; rupees at the VAN farmer price sheet, Aug 2026. 5 litres is where the 20 L pack starts to pay.

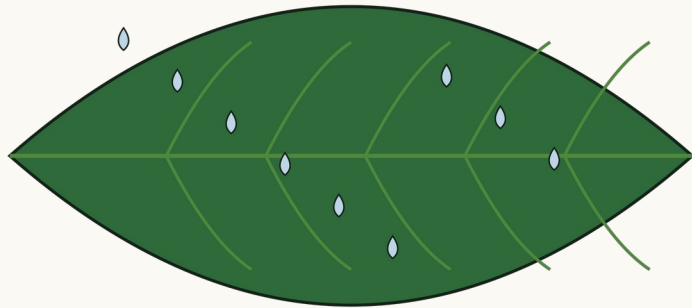
Crop	Stage and method, as printed	Litres	Zinc	Rupees per acre
Potato	Land preparation to maturity, spray, 5 rows	5	250 g	6,250
Chili	Transplanting to after picking, spray, 5 rows	5	250 g	6,250
Guava	October to February, foliar, 1 L per 100 L water, 5 rows	5	250 g	6,250
Banana, year 1	Early growth, fertigation, 1 to 2 L	1 to 2	50 to 100 g	1,250 to 2,500

Potato, chili and guava spray 1 litre at each of 5 rows across the season: 250 g of zinc and 6,250 rupees an acre in 1 L bottles, or 5 litres out of a 20 L pack at 875 a litre. Guava is the 1 programme that prints the water volume.

VAN's published crop programmes on van.com.pk, rows as printed. Rupees = litres × 1,250 at the 1 L price, farmer price sheet, Aug 2026; the 20 L pack is 17,500.

HOW IT GOES ON

On the leaf. Water first, then the bottle. Morning or evening. Never with the phosphate.



Sprayed onto the leaf. Small doses, taken up through the leaf surface, for a nutrient the crop needs quickly or that the soil locks up. Spray early morning or late afternoon; never on a wilted crop.

Water first, product second

Fill the tank with 100 to 200 litres of clean water an acre, then add the bottle, and keep it agitated. A tank that settles sprays the last acre at a different rate from the first. The dilution is part of the rate: cutting the water to save a pass doubles what lands on the leaf. The guava row prints its own water: 1 L per 100 L.

A fine, even fan, at the stage the plan prints

Cover the leaf without run-off. Morning or evening, never midday. 1 litre an acre at early growth on wheat and rice; at the rows the plan prints for potato, chili, pulses and guava. Banana and garlic run it through the water at the plan's rate.

Never in the same tank as a soluble phosphate

Zinc and a soluble phosphate make zinc phosphate, which does not dissolve: both nutrients are lost and the nozzle can block. Spray the micronutrients on their own pass. And never a drone rate from this deck: the drone grade is a different concentration with its own rate.

“1,250 for a litre? My zinc sulfate bag costs less than that.”

The dealer says, out loud

Write both on your paper. Then write what each is for. A bag of zinc sulfate is 1 metal, put in the soil, where above pH 8 the lime and the phosphate take most of it before the root arrives. This litre is 4 metals, 100 g together, on the leaf, where the soil never gets a vote, in 1 pass instead of 4. On wheat it is 1,250 rupees an acre, once, at early growth. What the crop does with it is not a number VAN prints for this product.

The fact behind it

- VL-Micromix 1 L is 1,250 rupees; the 20 L pack is 17,500 (farmer price sheet, Aug 2026).
- 50 g zinc, 20 g iron, 20 g manganese, 10 g copper in every litre (the declared analysis on the bottle).
- Wheat: 1 litre at early growth, sprayed (the wheat programme on van.com.pk).
- 69.1% of Punjab samples read above pH 8 (the Punjab soil record).

Never say: “It gives you X more maunds” or a rupees-per-acre return. No yield figure exists for the product. No cost per gram of zinc against the bag.

“My crop shows no shortage. Why would I spray metals it does not ask for?”

The dealer says, out loud

Because the shortage shows late. The 4 metals run the enzymes of growth, and a crop short of them makes itself small before the leaf shows anything. 59.0% of Punjab soil samples read low in zinc, and above pH 8 the soil locks iron and manganese with it. The programme puts 1 litre on at early growth for that reason. If you want to know your own field, a soil test tells you; the plan is written for the ground the survey describes.

The fact behind it

- 59.0% of 770,160 Punjab samples read below 1.0 ppm zinc; 69.1% read above pH 8 (the Punjab soil record).
- A zinc-short crop shows small, narrow young leaves, yellow between the veins, and short joints (VAN's 17-nutrients page).
- Wheat and rice take VL-Micromix at early growth (the crop programmes on van.com.pk).
- Copper reads Healthy in every Punjab district where the figure is comparable, 32 of 36 (the Punjab soil record).

Never say: “Every field is short of all 4.” Copper reads Healthy in every comparable Punjab district; VAN does not sell copper as a shortage.

“Can I put it in the tank with my DAP water, or with the phosphate spray?”

The dealer says, out loud

No. Zinc and a soluble phosphate in 1 tank make zinc phosphate, which does not dissolve. You lose the zinc, you lose the phosphate, and the nozzle can block. Spray the micronutrients on their own pass, and put the phosphate in the ground at land preparation where it belongs. The same reason is why heavy phosphate on a zinc-short soil makes the zinc shortage worse.

The fact behind it

- Zinc phosphate is poorly soluble: zinc and a soluble phosphate in 1 tank drop out together (textbook chemistry).
- Zinc and phosphorus pull against each other in the soil; heavy phosphate on a zinc-short soil makes it worse (VAN's 17-nutrients page).
- Every VL-Micromix row is its own pass: spray or fertigation, never a tank partner named (the crop programmes on van.com.pk).

Never say: “Stir it well and it is fine.”

“Will it burn the leaf?”

The dealer says, out loud

Not at the rate and the water the plan gives: 1 litre an acre in 100 to 200 litres of clean water. Burn comes from cutting the water to save a pass, and from spraying at midday. Water first, then the bottle, keep it agitated, a fine even fan, morning or evening. The guava row prints 1 L per 100 L of water; keep to it.

The fact behind it

- Guava: 1 L per 100 L of water, 5 sprays from October (the guava programme on van.com.pk).
- VAN's general spray guidance: 100 to 200 L of clean water an acre, water first, product second.
- Morning or evening, never midday (VAN's spray guidance).
- Cutting the water doubles the concentration on the leaf; the dilution is part of the rate (VAN's spray guidance).

Never say: A lower water volume, or “it cannot burn”.

“Can I put it in my drone?”

The dealer says, out loud

Not from this bottle at this rate. VAN flies a drone grade of VL-Micromix, and the site says so, but a drone tank is 20 litres and the concentration on the leaf is a different specification with its own rate. Ask VAN for the drone rate; I will not read one off a knapsack bottle.

The fact behind it

- A drone and ultra-low-volume grade of VL-Micromix is in production under the same name (the application-systems page on van.com.pk).
- A drone tank carries about 20 litres; everything the crop gets that day is dissolved in it (the same page).
- The knapsack rate is 1 litre in 100 to 200 litres of water; the drone grade has its own rate, from VAN (VAN's spray guidance).

Never say: “Use the 1 L bottle in the drone at knapsack strength.” No drone rate from this deck.

“Which stage, and how many bottles?”

The dealer says, out loud

What the plan prints. Wheat, basmati and hybrid rice: 1 litre at early growth, sprayed. Chickpea, lentil, mungbean and sunflower: 1 litre at grand growth. Potato and chili: 1 litre at each of the 5 rows across the season. Guava: 1 litre in each of the 5 months from October, at 1 L per 100 L of water. Banana and garlic go through the water. Never more than the row: more litres is more metal than the crop asked for, and zinc in excess pushes out iron.

The fact behind it

- 16 programmes name VL-Micromix, 40 rows, 1 litre per application on every row (the crop programmes on van.com.pk).
- Wheat, basmati, hybrid rice and sesame: early growth; chickpea, lentil, mungbean, sunflower: grand growth (the same programmes).
- Potato, chili, guava, soybean, strawberry and turmeric: 5 rows each across the season (the same programmes).
- Iron and manganese compete, and too much of 1 metal reduces another (VAN's 17-nutrients page).

Never say: “Double the bottles if the soil is very short”, or a rate for a crop with no row.

“My soil test says only zinc is low. Why pay for 4 metals?”

The dealer says, out loud

Then buy the zinc alone. VAN makes V-Zinc 10% for exactly that case, and VL-Boron for a boron shortage. VL-Micromix is for the field that has not been tested, or reads low on more than 1 metal, because on calcareous ground zinc, iron and manganese are locked by the same pH and usually go short together. 1 pass, 4 metals, ratios already set. The soil test decides; the dealer follows it.

The fact behind it

- V-Zinc 10% (zinc sulfate, 10% w/v) and VL-Boron (boron 5% w/v) are separate VAN products with their own rows (the products page on van.com.pk).
- Where a soil test shows 1 dominant shortage, the single-nutrient product is the better answer (VAN's own product guidance).
- Iron and manganese are locked by high pH and lime, the same condition that locks zinc (VAN's 17-nutrients page).

Never say: “Micromix does everything the single products do.” Each has its own rate and its own row.

“Is it registered? What is the licence?”

The dealer says, out loud

Pakistan Standard PS 5469-2020 for liquid nutrient supplements, manufacturing licence CM/L-3129/2024, VAN as the registered maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Read the batch number off the bottle and VAN sends you that batch's certificate, free.

The fact behind it

- VL-Micromix: PS 5469-2020, manufacturing licence CM/L-3129/2024 (the regulatory table on van.com.pk, from the PSQCA master record).
- The licence number is printed on the pack; the number is what a dealer quotes, never an expiry date.
- Every batch is released through VAN Lab, PNAC LAB 336, ISO/IEC 17025:2017 (VAN's quality page).

Never say: An expiry date. The number is what a dealer quotes.

VL-Micromix in the published programmes, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Early Growth	1 pack	Spray	1 L
Basmati Rice	Early Growth	1 pack	Spray	1 L
Hybrid Rice	Early Growth	1 pack	Spray	1 L
Potato	Land Preparation	1 pack	Spray	1 L
Potato	Germination	1 pack	Spray	1 L
Potato	Early Growth	1 pack	Spray	1 L
Potato	Grand Growth	1 pack	Spray	1 L
Potato	Maturity	1 pack	Spray	1 L
Chili	Land Preparation/Transplantation	1 pack	Spray	1 L
Chili	Germination	1 pack	Spray	1 L
Chili	Early Growth	1 pack	Spray	1 L
Chili	Grand Growth	1 pack	Spray	1 L

Rows as published in van.com.pk's crop programmes, 12 of the 16 programmes that name VL-Micromix. Spray on every row but garlic; 1 litre per application throughout; guava prints the water volume. A rate is per acre; the crop page scales it to the field.

VL-Micromix in the published programmes, continued.

Crop	Stage	Rate per acre	Method	Pack
Chili	After Picking	1 pack	Spray	1 L
Garlic	Maturity	1 pack	Fertigation	1 L
Chickpea	Grand Growth	1 pack	Spray	1 L
Lentil	Grand Growth	1 Pack	Spray	1 L
Mungbean & Mash	Grand Growth	1 Pack	Spray	1 L
Sesame	Early Growth	1 pack	Spray	1 L
Sunflower	Grand Growth	1 pack	Spray	1 L
Guava	October, flowering starts	1 pack	Foliar, 1 L per 100 L water	1 L
Guava	November, fruiting	1 pack	Foliar, 1 L per 100 L water	1 L
Guava	December	1 pack	Foliar, 1 L per 100 L water	1 L
Guava	January	1 pack	Foliar, 1 L per 100 L water	1 L
Guava	February. Harvest	1 pack	Foliar, 1 L per 100 L water	1 L

Rows as published in van.com.pk's crop programmes, 12 of the 16 programmes that name VL-Micromix. Spray on every row but garlic; 1 litre per application throughout; guava prints the water volume. A rate is per acre; the crop page scales it to the field.

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1 A yield figure, a “% more” or a hidden-hunger loss figure. No VL-Micromix trial exists; the demand-window argument is mechanism only.

3 That copper is a general shortage in Pakistan. It reads Healthy in every comparable Punjab district. It is in the bottle for the crop’s own need.

5 A drone rate, a fertigation rate the plan does not print, or a water volume other than the plan’s, as if it were a programme figure.

2 That 64% of Pakistan’s soils are zinc-deficient, or 19% iron, 4% manganese, 2% copper, until a source is on file. The Punjab survey’s 59.0% low in zinc is the figure a dealer may say.

4 That the 4 metals are chelated. VAN’s product data says “held in solution”, and the deck says no more than that.

6 That it replaces a soil test, or corrects a shortage a spray at flowering can no longer reach.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a certificate for VL-Micromix shows

The declared analysis tested for the batch: zinc, iron, manganese and copper against Zn 5 · Fe 2 · Mn 2 · Cu 1, % w/v, with the batch number, the Pakistan Standard PS 5469-2020 and licence CM/L-3129/2024 on the bottle. No certificate for a VL-Micromix batch was on file for this deck, so no batch reading is printed here; the grower asks for his own batch's certificate.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

The small nutrients. VAN has no promise word for them; a plain line.

Urdu first pass, unreviewed

چٹکی بھر غذا، فصل اس کے بغیر پوری نہیں ہوتی

Your soil has the metal and locks it. The leaf takes it straight in.

Urdu first pass, unreviewed

زمین میں دھات ہے مگر جکڑی ہوئی ہے، پتا اسے سیدھا لے لیتا ہے

1 litre an acre, water first, then the bottle. Morning or evening.

Urdu first pass, unreviewed

ایک لیٹر فی ایکڑ، پہلے پانی پھر بوتل۔ صبح یا شام

Never in the same tank as phosphate.

Urdu first pass, unreviewed

فاسفیٹ کے ساتھ ایک ٹینک میں کبھی نہیں

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

VL-Micromix: 4 metals on the leaf, 1 litre an acre, at the row the plan prints. Never with the phosphate.

